

InsightFlow

Data Cleaning and Analytics System

System Documentation (FEDM)

Course: FEDM — Fundamentals of Enterprise Data Management

Date: May 2026

Platform: Laravel 13 + React 19 (Inertia.js v3)

1. Introduction

InsightFlow is a web-based Data Cleaning and Analytics System that allows users to upload raw datasets in CSV or Excel format, apply automated and manual data cleaning operations, generate statistical insights, and create interactive visualizations. The system provides a complete end-to-end data processing workflow through an intuitive browser-based interface that requires no programming knowledge.

The purpose of this project is to demonstrate practical data preprocessing and analysis techniques by building a real-world application that addresses common data quality issues. The system automates the detection and correction of problems such as missing values, duplicate records, inconsistent formatting, and invalid entries, while providing clear explanations for every action taken.

The target users are students, researchers, analysts, and small business owners who need to clean and understand tabular datasets without writing code. The system is fully responsive and accessible through any modern web browser on both desktop and mobile devices.

2. Problem Statement

Raw data collected from surveys, databases, spreadsheets, and business systems is rarely clean or ready for analysis. Studies show that data professionals spend 60–80% of their time on data preparation rather than actual analysis. The most common data quality issues include:

- **Missing values** — Empty cells that reduce sample size and introduce bias in analysis results.
- **Duplicate records** — Repeated rows that inflate counts and distort frequency distributions.
- **Inconsistent formatting** — The same value appearing as "New York", "new york", and "NEW YORK", causing fragmented grouping.
- **Invalid data** — Impossible values like negative prices or ages over 150, resulting from data entry errors.
- **Outliers** — Extreme values that disproportionately affect averages and statistical models.
- **Wrong data types** — Numbers stored as text, preventing mathematical operations.

InsightFlow solves these problems through automated detection, intelligent cleaning algorithms, and transparent reporting that explains why each correction was made.

3. System Overview

InsightFlow combines data ingestion, profiling, cleaning, analysis, and visualization into a single application. Key features include:

Feature	Description
Data Upload	Import CSV/Excel files (up to 10MB, multiple files at once)
Data Profiling	Auto-detect data types, missing values, and column statistics
Quality Score	0–100% score measuring completeness, consistency, validity, uniqueness
Manual Cleaning	7 configurable operations (remove duplicates, fill missing, trim, etc.)
Auto-Clean	9-step intelligent pipeline with explanations for each action
Comparison	Side-by-side original vs. cleaned data with highlighted changes
Analysis	Statistics, trends, correlations, anomaly detection, interpretations
Dashboard	Unified view with charts, quality scores, insights, and trends
Visualization	6 chart types with filters, aggregation, and drill-down
Merging	Join two datasets using INNER, LEFT, RIGHT, or FULL joins
Export	Download cleaned data as CSV/Excel; reports as PDF
Sample Data	Built-in sample datasets for demonstration

4. System Workflow

Upload ? Profile ? Clean ? Analyze ? Visualize

Upload: Users upload CSV or Excel files. The system parses headers, detects metadata, and automatically names the dataset after the file. Multiple files uploaded together are linked as related datasets.

Profile: The system generates column-level statistics including data type, missing count, unique count, and numeric measures (min, max, average). A Data Quality Score (0–100%) is calculated based on completeness, consistency, validity, and uniqueness.

Clean: Users choose Manual Cleaning (configure specific operations) or Auto-Clean (9-step automated pipeline). Every action is logged with an explanation of what was done and why.

Analyze: The system calculates summary statistics (mean, median, std dev), detects trends, computes Pearson correlations between numeric columns, identifies anomalies via Z-scores, and generates English-language interpretations.

Visualize: Users create interactive charts by selecting columns, chart types (bar, line, area, pie, donut, scatter), and aggregation modes. The Dashboard page combines all elements in a single view.

5. Data Profiling

When a dataset is uploaded, InsightFlow generates a comprehensive profile to help users understand their data before cleaning.

Data Type Detection: The system samples up to 20 values per column and classifies each as number (>80% numeric), boolean (>80% yes/no/true/false), date (>80% parseable dates), or text (default). This determines which cleaning and analysis methods are applied.

Missing Value Identification: Each column's empty/null cells are counted and displayed as color-coded progress bars — green for >80% complete, amber for 60–80%, red for 60%.

Summary Statistics: Numeric columns show min, max, average, and sum. All columns show total count, missing count, and unique value count.

Data Quality Score: A composite 0–100% metric averaging four dimensions:

Metric	Calculation
Completeness	$(\text{filled cells} \div \text{total cells}) \times 100$
Consistency	% of text columns with uniform casing
Validity	% of values passing domain rules (age 0–150, price ? 0)
Uniqueness	$(\text{unique rows} \div \text{total rows}) \times 100$

6. Data Cleaning Methods

Manual Cleaning provides 7 operations: remove duplicates, remove missing rows, fill missing values, convert data types, trim spaces, standardize capitalization, and filter invalid data. Users configure each operation with specific columns and parameters.

Auto-Clean Engine applies 9 steps in optimized order:

#	Step	Why It's Applied
1	Data inspection	Identifies the scope of cleaning needed
2	Remove rows >50% missing	Mostly empty rows cannot be reliably imputed
3a	Context-based imputation	Infers missing values from correlated columns
3b	Mean/mode fill	Mean for numbers, mode for categories preserves distribution
4	Remove duplicates	Duplicates inflate counts and skew analysis
5	Type conversion	Enables math operations on numeric text
6	Standardization	Trim spaces + normalize casing for consistent grouping
7	IQR outlier capping	Limits distortion from extreme values
8	Filter invalid data	Removes impossible values (negative prices, age >150)
9	Computed columns	Generates Total = Price × Quantity when applicable

7. Insights Generation

After cleaning, InsightFlow generates statistical insights automatically:

Summary Statistics: For each numeric column — count, mean, median, standard deviation, min, max, range, and sum.

Trend Detection: Compares the average of the first half vs. second half of values. Reports whether the trend is increasing, decreasing, or stable with percentage change.

Frequency Analysis: Identifies the top 5 most frequent values per column with occurrence counts and percentages.

Correlation Analysis: Calculates Pearson correlation coefficient (r) between all pairs of numeric columns. Results classified as Very Strong (>0.8), Strong (>0.6), Moderate (>0.4), Weak (>0.2), or Very Weak (<0.2).

Anomaly Detection: Flags values with Z-scores exceeding ± 3 standard deviations as statistical outliers.

English Interpretations: All findings are translated into readable sentences, e.g., "Strong positive correlation between 'Price' and 'Total' ($r = 0.95$)" and "Column 'Revenue' shows an increasing trend (23.5% change)."

8. Dashboard & Visualization

Chart Type	Best Used For	Example
Bar Chart	Comparing values across categories	Sales by product category
Line Chart	Trends over sequential data	Monthly revenue over a year
Area Chart	Cumulative totals and volume	Cumulative customer growth
Pie Chart	Proportions of a whole	Market share distribution
Donut Chart	Proportions with center summary	Budget allocation with total
Scatter Plot	Relationships between two variables	Price vs. quantity patterns

These six types cover the four fundamental visualization categories: comparison (bar), trend (line/area), composition (pie/donut), and relationship (scatter).

Interactive features: Column selection, 5 aggregation modes (sum, avg, count, min, max), tooltips, drill-down, and automatic dark mode adaptation.

Dashboard page: Combines quality score ring, stat cards, bar and line charts with selectors, key insights list, correlation heatmap, and trend indicators — all on one page.

9. System Implementation

Layer	Technology	Version
Backend	Laravel (PHP 8.4)	v13
Frontend	React + TypeScript	v19
SPA Bridge	Inertia.js	v3
Styling	Tailwind CSS	v4
Charts	ApexCharts	v5
Animations	Framer Motion	v12
Auth	Laravel Fortify (2FA)	v1
PDF Export	DomPDF	v3
File Import	Maatwebsite Excel	v3
Testing	Pest PHP (97 tests)	v4
Build Tool	Vite	v8

Architecture: Monolithic Laravel backend with Inertia.js serving React pages. Dataset content stored in JSON columns for schema flexibility. All data processing runs server-side. Top-bar navigation provides a guided workflow. Fully responsive with mobile hamburger menu.

10. Sample Dataset & Output

InsightFlow includes two built-in sample datasets loaded via a "Load Sample Data" button:

Sales Data (15 rows, 6 columns): Contains products, categories, prices, quantities, dates, and regions — with intentional duplicates, missing values, inconsistent casing, negative prices, and empty rows.

Student Grades (12 rows, 6 columns): Contains names, ages, subjects, scores, grades, and semesters — with duplicates, invalid ages, missing scores, and casing inconsistencies.

Auto-Clean Results (Sales Data)

- 1 empty row removed (>50% missing)
- 1 duplicate row removed
- Missing tablet price filled with column mean
- Casing normalized: "mouse" ? "Mouse", "electronics" ? "Electronics"
- Spaces trimmed: " Laptop " ? "Laptop"
- Negative price row removed (Phone at -\$50)
- Total column generated ($\text{Price} \times \text{Quantity}$)
- **Quality Score:** ~65% ? ~95%

11. Limitations

- **File size limit:** Maximum 10MB per file. Larger datasets require preprocessing.
- **Tabular data only:** Supports CSV and Excel only. No JSON, XML, or unstructured data.
- **No machine learning:** Uses statistical methods only. No predictive modeling or clustering.
- **Static datasets:** No live database connections or real-time data streaming.
- **Basic imputation:** Mean/mode fill only. No KNN or regression-based imputation.
- **English only:** Interface and generated interpretations are in English.
- **All-or-nothing auto-clean:** Cannot selectively enable/disable individual auto-clean steps.
- **Modern browser required:** Charts and animations need Chrome, Firefox, Edge, or Safari.

12. Recommendations / Future Improvements

- **Advanced imputation:** Add KNN and regression-based methods for more accurate missing value estimation.
- **Reusable cleaning recipes:** Save cleaning configurations as templates for repeated use.
- **REST API:** Provide programmatic access for integration with other tools.
- **Larger file support:** Chunked processing and background jobs for files over 10MB.
- **Custom validation rules:** User-defined domain rules (email format, phone patterns).
- **Pivot/unpivot:** Reshape data between wide and long formats.
- **Real-time collaboration:** Multiple users viewing and annotating the same dataset.
- **AI-powered insights:** LLM-generated natural-language analysis summaries.
- **Scheduled pipelines:** Recurring automated cleaning for regular data uploads.

13. Screenshots

Capture and insert the following screenshots from the running application:

Screenshot 1 — Landing Page: The welcome page at /welcome with hero section, workflow pipeline, feature cards, and call-to-action buttons.

Screenshot 2 — Upload Page: File upload form, dataset list, feature overview cards, and "Load Sample Data" button.

Screenshot 3 — Data Profile: Overview stat cards, Data Quality Score ring with sub-metrics, column statistics table, and missing values progress bars.

Screenshot 4 — Cleaning Interface: Manual cleaning operation builder with column selectors and the Auto-Clean button.

Screenshot 5 — Data Comparison: Before/after quality scores, cleaning log with "Why" explanations, side-by-side tables with highlighted changes, export buttons.

Screenshot 6 — Analysis Results: Summary statistics, trend indicators, frequency distributions, correlations, anomalies, and English interpretations.

Screenshot 7 — Dashboard: Quality score ring, stat cards, interactive charts, insights list, correlation heatmap, and trend indicators.

Screenshot 8 — Visualization Builder: Rendered chart with column selector, chart type switcher, aggregation options, and interactive tooltips.